Toxic Language Captures Attention: Evidence from 200 Million News Headline Impressions

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Online platforms extract revenue by capturing user attention and selling it to advertisers, leading algorithmic content curation to prioritize content that most effectively captures attention. This creates a potential pathway for toxic language to dominate public discourse if toxic content systematically outperforms civil alternatives in capturing clicks. Using data from 53,755 news headlines across 12,473 A/B tests that generated over 205.8 million impressions at Upworthy.com, I employ the Perspective API to measure headline toxicity and analyze click-through rates as the primary outcome variable to approximate attention capture. A fixed effects specification exploits within-experiment variation to isolate toxicity effects from article-level characteristics, comparing headlines that advertise identical content but vary in toxic language. Results demonstrate that toxic content increases click-through rates by 9.859% relative to civil alternatives, even when controlling for negativity and other linguistic features. This persistent toxicity bias suggests that toxic language triggers automatic attention-capture mechanisms that operate free from social signaling dynamics.

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"You affect the world by what you browse."
— *Tim Berners-Lee (attributed, source uncertain)*

Introduction

Attention has become the fundamental resource that online platforms extract and monetize (Loewenstein and Wojtowicz 2025; Berger, Moe, and Schweidel 2023). Despite remarkable technological developments in media platforms (from newspaper journalism and radio to television, streaming, podcasts and social media), the underlying business model remained remarkably stable: ad-supported platforms provide free content to capture the users' attention, then sell that attention to advertisers. To achieve this, platforms curate their content inventory and typically embed paid advertisements in ways that are difficult to ignore (Wu 2017). This model evolved as user-generated content expanded content inventories and required automated content delivery systems that maximize attention and advertisement revenue (Aridor et al. 2024). Where traditional media outlets employed human editors to curate content, modern platforms deploy data-driven recommender systems that determine which content is displayed to whom. However, these algorithms amplify whatever content attracts the most attention, regardless of its social value: since users are often drawn to content that is not only engaging but also toxic (e.g., hate speech, harassment, and related forms of content, see Beknazar-Yuzbashev, Jiménez-Durán, and Stalinski 2024), platforms may inadvertently promote content that degrades public discourse and the users' well-being (Guess et al. 2023; Bursztyn et al. 2023).

Analyzing 53,755 news headlines in a unique setting of 12,473 A/B-tests that generated over 205.8M impressions, I find that toxic content systematically outperforms civil alternatives in capturing clicks. Specifically, toxic content increases click-through rates (CTR) by 9.859% even when advertising identical articles.

These findings reflect deeper psychological mechanisms that characterize the platforms' "battle for clicks and eyeballs" (Senior 2016). In competitive information environments where each piece of content only has (milli-) seconds to attract a user's attention, content can be curated to stand out (e.g., because of visual contrasts, social and personal significance, or emotional significance) and capture attention before any deliberate evaluation occurs. Hence, platforms can leverage so-called bottom-up mechanisms that operate outside of a user's volitional

control (Loewenstein and Wojtowicz 2025) and arise in response to properties of their stimuli to capture and, ultimately, steer their users' attention.

Evidence from multiple disciplines suggests that negative content (Robertson et al. 2023; Banerjee and Urminsky 2024) may be particularly effective at capturing attention through these automatic processes. Baumeister et al. (2001) suggest that an attraction to such content may be deeply rooted due to evolutionary origins: organisms that were better attuned to bad things would have been more likely to survive threats (see also Öhman and Mineka 2001). Accordingly, toxic language may trigger threat-detection mechanisms even when the "threats" are manufactured for a platform's monetization rather than a user's survival (Rathje and Van Bavel 2025).[1]

Prior literature on toxicity not only found negative effects on welfare but also positive effects on engagement (Beknazar-Yuzbashev et al. 2022; Beknazar-Yuzbashev, Jiménez-Durán, and Stalinski 2024). However, conceptualizing engagement faces a methodological challenge: to disentangle toxicity's direct effect on individual content consumption from confounding social dynamics. Because such evidence comes from social media platforms, the reported engagement decisions were public and may have served social signaling functions (Huang and Lin 2025) rather than information seeking. To address this concern, I analyze private click decisions in a field experimental setting where users click on headlines solely to access article content, with no immediate social visibility or signaling opportunities.

The remainder of this paper proceeds as follows. The next section describes the Upworthy Research Archive dataset, explains the toxicity measurement approach using the Perspective API, and discusses click-through rates as the primary outcome variable. Next, I outline the empirical strategy that exploits within-experiment variation to identify the association between toxicity and engagement. I then present main findings and robustness checks, followed by a discussion of the implications for content optimization and platform design. The paper concludes with directions future research.

[1] In fact, recent studies suggest that there is a disconnect between the users' and the media platforms' welfare: while Beknazar-Yuzbashev et al. (2022) show that toxic content may be profitable, other experimental evidence also suggests that active social media users would prefer platforms such as TikTok and Instagram not to exist (Bursztyn et al. 2023). Rathje and Van Bavel (2025) term this disconnect the *"paradox of virality"* and compare the engagement with toxic content to rubbernecking at car accidents. People feel compelled to look even when they consciously wish to look away.

Data

The Upworthy Research Archive

This paper analyzes variations of news headlines written by writers and editors of Upworthy.com, a U.S. media publisher that was recognized as the fastest-growing media company in the world in 2013. Upworthy pioneered the systematic use of randomized A/B-tests (Matias et al. 2021), hereafter referred to as *"experiments"*. For each article, Upworthy tested different variations of headlines to advertise the respective article on its home feed. The corresponding experimental process was straightforward: editors or writers created several alternative headlines per news article, website visitors were randomly assigned to one version upon visiting Upworthy.com, and the platform recorded whether the visitor clicked, producing a binary outcome for each headline. The headline with the most clicks per impression (referred to as click-through rate, CTR) was then selected to advertise the article permanently and for all users. Because these experiments were not only conducted systematically, but also deployed at scale (i.e., at a time when Upworthy's content was shared more frequently on Facebook than that of all major U.S. outlets combined) this case is well-suited to study attention allocation and data-driven content optimization.

The *Upworthy Research Archive* (Matias et al. 2021) retrieved the corresponding data and contains all experiments conducted between January 24, 2013, and April 30, 2015. For this study, I use the cleaned version of the dataset provided by Reiss and Roggenkamp (2025), who removed duplicate headlines and added additional attributes (e.g., semantic negativity scores predicted by gpt-4o-mini). The resulting subset includes 12,473 experiments with 53,755 unique headline variations, that generated approximately 205.8M impressions and 2.8M clicks.

The dataset has a two-level structure: the article level corresponds to an experiment (A/B-test), and the headline level, where multiple alternatives advertise the same article. Table 1 displays six alternative headlines that advertised the same underlying article to illustrate an experiment's nested data structure. The headlines exhibit substantial variation in both toxicity scores (ranging from .044 to .859) and click-through rates (ranging from .343% to .866%).

In contrast to online platform and observational studies (see Roggenkamp, Boegershausen, and Hildebrand 2025), the data were not affected by algorithmic interference issues (see, e.g., Xu, Zhang, and Zhou 2020; Johnson 2023; Braun and Schwartz 2025). In addition, Upworthy conducted only one experiment at a time across their entire website to ensure that the tests were independent.

Matias et al. (2021) collaborated and interviewed with former engineers, editors, and data scientists employed at Upworthy to document and validate the dataset structure. All seven interviewees confirmed details about the experimental process and data collection procedures. An additional quantitative validation checked balance across experimental arms and confirmed that clicks were always lower than or equal to participant counts.[2]

Table 1: Headlines from an examplary A/B-test describing the same article

Headline	Toxicity	CTR (%)
Here's How You End Up In Prison If You're British And Also A Complete Idiot	0.859	0.866
He's The Dumb Husband. She's A Doting Wife. And Then Something Important Landed At Their Door.	0.627	0.674
What They Learned Will Change What They Think About The Planet. (But They'll Still Be British.)	0.112	0.343
In Just 1 Minute, These Two People Learned The One Comment They're Not Supposed To Make	0.109	0.596
A Couple That Will Never Make The Same, Simple Mistake Again	0.106	0.493
An Incredibly Important Piece Of Information Came Right To Their Front Door	0.044	0.350

Note: Toxicity scores range from 0 to 1 and represent the probability that human raters would classify the content as toxic, as measured by the Perspective API. CTR = Click-Through Rate (clicks/impressions). Headlines are ordered by toxicity score (highest to lowest). All headlines advertised the same underlying article.

Toxicity Prediction

To measure toxicity in the Upworthy headlines, I employed the Perspective API (Lees et al. 2022; Votta 2025), a machine learning system developed by Jigsaw (a unit within Alphabet Inc.) for identifying toxic content in online conversations. The API was released in 2017 and has been widely adopted by major publishers including The New York Times, Le Monde, and The Financial Times for content moderation purposes.

[2]The validation process identified treatment imbalance issues for tests conducted between June 25, 2013, and January 10, 2014, leading to the exclusion or special treatment of experiments from this period in subsequent analyses.

The underlying models are trained on millions of comments from a variety of sources, including Wikipedia and The New York Times, across several languages. For each comment, 3-10 human annotators evaluated whether the comment contains toxicity following standardized instructions. As a result, the API generates toxicity scores on a continuous scale from 0 to 1, that represent the probability that a given text would be perceived as toxic by human raters. For this reason, the scores have a direct interpretation based on the training methodology: a score of 0.7 suggests that 7 out of 10 raters would classify content as toxic. The API's definition of toxicity, therefore, is rooted in the training instructions provided to human annotators. Specifically, they evaluated the toxicity in each comment, with "toxic" content defined as "a rude, disrespectful, or unreasonable comment that is somewhat likely to make you leave a discussion or give up on sharing your perspective" (Lees et al. 2022). For the analysis, I processed each headline in the Upworthy dataset through the Perspective API to obtain toxicity scores. The API employs character-level transformer models that are well-suited for analyzing news headlines, as they can handle the varied linguistic styles and creative language common in attention-grabbing content. With an Area Under the Curve (AUC) of .974 for toxicity detection (Lees et al. 2022), the API provides both strong empirical performance in addition to its ecological validity (given that it is deployed by major publishers for real-world content moderation).

Despite predicting continuous likelihood scores, I also use the predictions to flag (and classify) headlines that exceed a threshold of .700. This threshold represents a critical benchmark in content moderation practices that typically triggers moderation actions such as reduced visibility or removal (see also Horta Ribeiro, Cheng, and West 2023). I thereby follow the Perspective API's recommendation[3] to use thresholds between 0.7 and 0.9 to mirror typical industry moderation use cases.

The predicted toxicity scores range from .000 to .929. Yet, only 295 (.623%) reached the recommended minimum threshold of .700. Hence, the corresponding distribution is right-skewed as displayed in Panel A of Figure 1. The scarcity of toxic headlines likely reflects the editorial and reputational constraints of a professional media outlet. As a result, the claims derived from this dataset may best be extrapolated to other, potentially user-generated contexts, that a characterized by predominantly civil discourses.

Even though the Perspective API represents state-of-the-art technology for automated toxicity detection, several limitations must be acknowledged. The models may be sensitive to

[3] https://developers.perspectiveapi.com/s/about-the-api-score, accessed 2025-08-12

the presence of profanity or certain keywords, potentially flagging content that uses strong language but is not necessarily harmful in context. Additionally, the API's performance varies across different types of content and may exhibit biases related to the demographic characteristics of the original training annotators or the specific domains represented in the training data. However, for research purposes examining patterns of toxicity in news headlines, the API provides a consistent measurement approach that enables large-scale analysis while maintaining interpretability through its probabilistic framework.

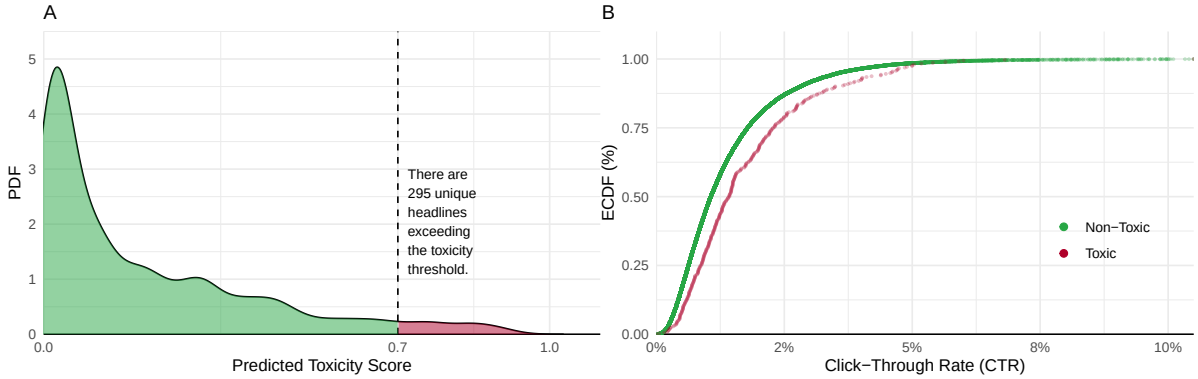
Click-Through Rate

Throughout this paper, the click-through rate (CTR), calculated as the ratio of clicks to impressions for each headline variation, is the primary dependent variable. Because users click on a headline to reveal more information related to that headline, the CTR offers several advantages for measuring attention allocation. First, a click represents behavioral outcome that requires minimal but non-trivial effort. As a consequence, the CTR captures users' intention to acquire information rather than passive exposure. Second, clicks are consequential as they reflect trade-offs with opportunity costs: choosing to learn more about one headline means forgoing alternatives (Reutskaja et al. 2011; Schütz and Stewart 2025). Third, the binary nature of clicks aligns well with the research question about whether toxic language influences attention capture. I am specifically interested in the initial (bottom-up) decision to acquire more information (i.e., the moment where headline characteristics translate into user behavior) rather than subsequent reading time or content consumption (Berger, Moe, and Schweidel 2023). Finally, clicks also represent the actual optimization metric that Upworthy used to guide editorial decisions. This is aligned with the information recommender systems process and implies that the findings speak directly to the incentives and trade-offs content creators and platform algorithms face more generally.

Overall, the recorded CTRs range from .000% to 14.891% and the distribution of click-through rates is right-skewed: most headlines attract relatively little attention, but a small fraction achieve high CTRs.[4] This long-tail distribution (see Panel B in Figure 1) mirrors patterns of virality in digital media, where a few pieces of content disproportionally attract attention. It also means that effect sizes expressed in percentage points should be interpreted relative to a low baseline median CTR of 1.067%.

[4]Only .072% of all headlines generated CTRs above 10%.

Figure 1: Distribution of predicted toxicity scores and click-through rate performance



Distribution of Predicted Toxicity Scores Across Headlines, A. The density plot shows that most headlines receive low toxicity scores, with only 295 headlines (.623% of the sample) exceeding the .700 threshold for toxic content. **Empirical Cumulative Distribution of Click-Through Rates by Toxicity Classification, B.** Headlines exceeding the toxicity threshold (red) systematically achieve higher click-through rates than non-toxic alternatives (green), with the toxic distribution shifted rightward across the entire range of performance outcomes.

Empirical Strategy

Panel B of Figure 1 provides initial, model-free evidence of a relationship between toxicity and engagement. The empirical cumulative distribution functions illustrate that headlines flagged as toxic consistently achieve higher click-through rates than civil alternatives, with the toxic distribution shifted to the right. This raw pattern suggests that toxic content captures more attention across the entire distribution of the CTR. However, this model-free comparison may be misleading if toxic headlines systematically advertise different types of articles than non-toxic headlines. Articles that naturally attract toxic language (such as controversial political topics) might also inherently generate higher CTRs regardless of headline phrasing. To better capture the effect of toxic language from these article-level characteristics, I exploit variation in headline toxicity within experiments to estimate whether CTR increases in toxicity when comparing headlines that advertise identical content.

To this end, I employ a fixed effects specification that controls for all article-level characteristics:

$$\text{CTR}_{ij} = \alpha_i + \beta \cdot \text{Toxicity}_{ij} + \gamma' \mathbf{X}_{ij} + \varepsilon_{ij}$$

where CTR_{ij} is the click-through rate for headline j in experiment i, Toxicity_{ij} is the

Perspective API's toxicity score (0-1), and α_i are experiment fixed effects. The vector $\mathbf{X}_{ij}$ includes headline-level controls: all models include headline length (word count) and text complexity (Gunning Fog Index) as covariates since Robertson et al. (2023) found these to predict CTR.[5]

The coefficient of interest, β captures the association between toxicity and CTR using only variation across headlines for the same article. For example, this approach compares the following two headlines advertising the same article reported in Table 1: *"An Incredibly Important Piece Of Information Came Right To Their Front Door"* with *"Here's How You End Up In Prison If You're British And Also A Complete Idiot"* (toxicity scores: .044 vs. .859) while holding constant the underlying article content and all article-invariant factors.

The remaining identification challenge is that editors have not systematically varried toxic language within their A/B-tests. Instead, variation in toxicity occurs naturally and may have been paired with other engagement tactics. If editors systematically bundled toxic language with specific engagement tactics within the same article, β captures the combined effect rather than toxicity's isolated impact. Banerjee and Urminsky (2024) note that in their review of the prior academic literature on linguistic drivers of engagement, academic experiments designed to test a single construct typically also included both intended and incidental differences because of inherent linguistic limitations that often required more changes than the intended change to preserve message realism (see also Simonsohn, Montealegre, and Evangelidis 2025) In addition, the scale of the dataset (which spans 12,473 experiments created by multiple editors over two years) exhibits heterogeneity in editorial approaches. This heterogeneity means toxic language is likely paired with different engagement tactics across experiments rather than systematically bundled with the same strategies, allowing β to better capture toxicity's effect on CTR.

Results

The baseline specification (see Model 1 in Table 2) reveals a positive association between toxicity and engagement: a one standard deviation increase in toxicity corresponds to a .045 percentage point increase in CTR (SE = .003, $z = 14.907$, 99% CI = [.037, .053]).

Model 2 tests whether toxicity's effect persists when controlling for negativity (as measured in Reiss and Roggenkamp 2025). The minimal change in toxicity's coefficient (from .045 to

[5] Platform age, another covariate they controled for in their mixed effect models, varies only between experiments, so article fixed effects absorb this variation.

.042) when adding negativity suggests that toxicity adds predictive power beyond negativity (SE = .003, z = 13.669, 99% CI = [.034, .050]). Consistent with this interpretation, the two measures are only moderately correlated ($\rho_{Pearson}$ = .242), which suggests partial overlap but also distinct variance. Users therefore appear drawn both to negative content (bad news, unfortunate events) and to hostile or disrespectful language, with these effects being additive rather than redundant. This pattern indicates that toxicity captures something beyond mere negativity, such as hostile, disrespectful, or threatening elements that may trigger automatic threat-detection responses. Importantly, both toxicity and negativity show, by and large, comparable positive associations with CTR, which underscores that each has a meaningfull effect on CTR.

Model 3 addresses a measurement concern noted in the Perspective API documentation: the algorithm may conflate profanity with toxicity, potentially misspecifying content that relies on strong language as toxic. As a consequence, there is a high correlation between toxicity and profanity ($\rho_{Pearson}$ = .865), which requires a careful interpretation. When both are included, toxicity's coefficient increases to .059 percentage points (SE = .006, z = 10.153, 99% CI = [.044, .074]) while profanity shows a negative association (b = -.019, SE = .005, z = -3.519, 99% CI = [-.032, -.005]). Rather than indicating that profanity reduces engagement, this pattern likely reflects the multicollinearity between these measures as the model cannot reliably separate their individual contributions. Hence, the most conservative interpretation focuses on Model 2, which suggests that toxic language, irrespective of the level negativity, is associated with higher engagement rates.

To contextualize these effect sizes: if the most toxic headlines in the dataset (scoring above 0.7) were compared to civil alternatives for the same articles, the model predicts approximately 9.859% higher click-through rates. However, this extrapolation assumes linearity across the full toxicity range, which may not hold given the limited variation in highly toxic content within this professionally edited dataset.

Results are robust to alternative modeling approaches (see Web Appendix A). As an additional robustness check, I estimated mixed effects binomial models that explicitly account for the binary nature of click decisions and include overdispersion correction to handle extra-binomial variation in the data. These models also include platform age as an explicit control variable to account for temporal changes in user engagement and editorial effectiveness over the study period, whereas the fixed effects models absorb temporal variation through experiment-level fixed effects. Platform age controls for the documented decline in headline performance

Table 2: Associations of Content Characteristics and Click-Through Rate

Model:	(1)	(2)	(3)
Variables			
Toxicity	0.045***	0.042***	0.059***
	(0.003)	(0.003)	(0.006)
Negativity		0.067***	0.061***
		(0.013)	(0.013)
Profanity			-0.019***
			(0.005)
Fixed-effects			
Experiment-Level	Yes	Yes	Yes
Fit statistics			
Observations	53,755	53,748	53,748
Pseudo R^2	0.704	0.705	0.705

Clustered (Experiment-Level) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05*

over time as users became accustomed to Upworthy's editorial practices (Robertson et al. 2023; Reiss and Roggenkamp 2025). The mixed effects specifications yield marginal effects of .049 percentage points, compared to .042 from the preferred fixed effects specification (Model 2).

Discussion

This research contributes to the emerging literature on toxic content and user behavior (Beknazar-Yuzbashev et al. 2022) by providing evidence that toxic language systematically captures more attention than civil alternatives, even when advertising identical content. Using a large-scale experimental dataset from Upworthy that isolates toxicity effects from social signaling concerns, I find that content reaching toxicity thresholds (where 7 out of 10 raters would classify it as toxic) is associated with a 9.859% increase in click-through rates.

These findings complement recent work on platform dynamics and user welfare. Whereas Guess et al. (2023) and Bursztyn et al. (2023) document how social media algorithms can create collective action problems that reduce user welfare, my results provide a mechanism for this phenomenon: toxic content exploits automatic attention capture processes that operate

below conscious control. This aligns with analyses of viral content (i.e., Berger and Milkman 2012), which found that affect-laden content drives engagement regardless of valence. While I cannot replicate their finding that positive valence drives virality (I find a negativity bias consistent with Robertson et al. 2023), I do replicate their broader point about emotional content.

The toxicity bias documented here has important implications for platform design and content moderation. Unconstrained recommender systems that optimize purely for attention metrics create systematic selection pressure favoring toxic content. My results demonstrate this holds true even in professional editorial contexts where toxicity is rare, and where the optimization algorithm (selecting headline variations with the highest click-through rates) is constrained. The findings generated in such a simple and manual, yet interpretable recommender system suggests that toxicity's attention-capturing effects are robust across different content environments and optimization approaches.

However, several methodological considerations warrant discussion. Hence, the interpretation of these findings requires careful consideration of what can and cannot be claimed from this observational design. While the within-experiment variation provides cleaner identification than cross-sectional comparisons, it must be acknowledged that this dataset captures editorial choices made in pursuit of engagement rather than scientific manipulation of toxicity: the variation in toxicity represents real-world optimization decisions where inflammatory language likely co-occurs with other attention-grabbing techniques.

Hence, this research suffers from the same limitation that is also present in many lab studies (Banerjee and Urminsky 2024; Simonsohn, Montealegre, and Evangelidis 2025): changing a single dimension while maintaining naturalistic text often proves impossible. In this light, the presented field evidence captures how toxicity operates in practice (i.e., bundled with whatever complementary tactics editors naturally employ). As a consequence, my results may be interpreted as the 'total effect' of using language that gets flagged as toxic in real-world content optimization, which may include both direct toxicity effects and correlated engagement tactics.

However, the scale and diversity of the used data offers reassurance. With thousands of independent editorial decisions spread across years and multiple writers and editors, it becomes implausible that toxicity consistently paired with the same hidden moderators. Moreover, the persistence of toxicity's effect after accounting for negativity indicates that the analyses are capturing something beyond simple valence.

Future research should address several theoretical and methodological questions raised by this study. Most pressing is the identification challenge where toxicity naturally co-occurs with other engagement tactics. While questions of external and (to a certain degree also internal) validity are inherently empirical and rarely resolved by single studies, a cumulative body of evidence across varied contexts could help disentangle whether the observed effects stem from co-occuring constructs such as arousal, surprise, norm violation, or toxicity per se.

Beyond identification, the functional form of toxicity's effect deserves attention. The current analysis assumes linearity, yet psychological responses to toxicity likely exhibit threshold effects or diminishing returns. Digital In-Context Experiments (Roggenkamp, Boegershausen, and Hildebrand 2025) could address this by continuously manipulating toxicity levels across entire feeds rather than examining single headlines in isolation. Such approaches could reveal whether toxicity operates through contrast effects and salience (where occasional toxic content stands out) or habituation (where sustained exposure reduces impact). Digital In-Context Experiments could also simulate various platform environments, from traditional news feeds to social media interfaces or oven review platforms to improve generalizability and to bridge the gap between this study, focusing on mass media feeds, and the prevailing social media studies.

The economic dimensions of toxic content also warrant systematic investigation. When Elon Musk announced plans to relax Twitter's content moderation policies in 2022, GroupM, the world's largest media buyer, classified Twitter as a "high-risk platform," leading many luxury brands like Balenciaga to pause or cease their advertising (Andres, Rossi, and Tremblay 2023). These reactions appear economically rational: research shows that advertisers bid more when they know their ads will appear in brand-safe contexts (Ada, Nabout, and Feit 2022), and premium brands particularly depend on high-quality content environments for successful campaigns (Shehu, Abou Nabout, and Clement 2021). This suggests that platforms face not just a binary trade-off between societal welfare and engagement, but must navigate a complex optimization problem where toxic content may drive user engagement while simultaneously reducing advertising revenue. Future research should formalize these competing pressures through game-theoretic models (see also Beknazar-Yuzbashev, Jiménez-Durán, and Stalinski 2024) and empirical studies comparing platform behavior before and after major advertiser boycotts across platforms.

Concluding Thoughts

This study provides evidence that toxic language captures attention through automatic processing mechanisms, creating incentives for content creators and recommendation algorithms to favor such content. While my analysis focuses on a specific platform (Upworthy.com) and content type (news headlines), the implications may extend to other attention-based contexts including email marketing, push notifications, and social media feeds. Understanding these dynamics is crucial for designing information environments that balance engagement with user welfare and discourse quality.

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Appendix A: Mixed Effects Models as Robustness Check

The main analysis employs fixed effects specifications that include experiment-level fixed effects (α_i) to control for all article-level characteristics. This approach exploits within-experiment variation by comparing headlines that advertise identical content but vary in toxic language. As this strategy holds constant the underlying article content, editorial context, and temporal factors, it absorbs all between-experiment variation that might be informative.

As a robustness check, I therefore estimate mixed effects binomial models that explicitly account for the binary nature of click decisions while allowing for both within- and between-experiment variation. These models include random intercepts for experiments to capture article-level heterogeneity and incorporate overdispersion correction to handle extra-binomial variation commonly observed in click data. The mixed effects approach also permits the inclusion of platform age as an explicit control variable, which accounts for temporal changes in user engagement and editorial effectiveness over the study period.

The mixed effects specifications take the form:

$$\text{logit}(\theta_{ij}) = \beta_0 + \beta_1 \text{Toxicity}_{ij} + \beta_2 \text{Negativity}_{ij} + \beta_3 \text{Profanity}_{ij} + \gamma' X_{ij} + u_i + \varepsilon_{ij}$$

where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ represents random intercepts for experiments.[6]

Table 3 presents results from three mixed effects specifications that mirror the fixed effects models in the main text. Model 1 includes toxicity and control variables, Model 2 adds negativity measured via GPT-4o semantic analysis, and Model 3 incorporates profanity scores. The models use impression weights and estimate marginal effects at the baseline click-through rate of 1.354%.

The mixed effects results support the robustness of the main findings. The toxicity coefficient in Model 2 yields a marginal effect of .049 percentage points, compared to .042 from the preferred fixed effects specification (Model 2 in Table 2). This consistency across modeling approaches strengthens confidence that the observed toxicity bias reflects genuine behavioral responses rather than specification-dependent artifacts.

[6] A dispersion parameter φ accounts for overdispersion beyond what the binomial distribution would predict through the variance specification: $\text{Var}(Y_{ij}) = \varphi \cdot n_{ij}\theta_{ij}(1 - \theta_{ij})$ where n_{ij} represents the number of impressions for headline j in experiment i.

Table 3: Mixed Effects Models: Toxicity and Click-Through Rate

	Model 1	Model 2	Model 3
Toxicity	0.053^{***}	0.049^{***}	0.071^{***}
	(0.001)	(0.001)	(0.003)
Negativity		0.082^{***}	0.075^{***}
		(0.005)	(0.005)
Profanity			-0.024^{***}
			(0.002)
Num. obs.	53755.000	53748.000	53748.000
AIC	531679.107	531383.797	531288.453

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

Note: Marginal effects in percentage points calculated at baseline CTR of 1.354%. Control variables (length, complexity, platform age) omitted for brevity but included in estimation. Random intercepts for experiments account for article-level heterogeneity. Overdispersion parameter estimated to handle extra-binomial variation. Models use binomial family with logit link and impression weights.